



National Institute of Business Management

## Robotic Application Development (Greenhouse Automation)

|         |   |                    |
|---------|---|--------------------|
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## Table of Contents

|                                                                          |   |
|--------------------------------------------------------------------------|---|
| Project Description .....                                                | 3 |
| Expected Functionalities .....                                           | 3 |
| 1. Soil Moisture Monitoring and Control .....                            | 3 |
| 2. Air Temperature / Humidity Control .....                              | 3 |
| 3. Greenhouse Efficiency (Sunlight) Monitoring .....                     | 3 |
| 4. Water Tank / Reservoir Level Monitoring .....                         | 4 |
| 5. Crop Watering based on Weather .....                                  | 4 |
| 6. Interactive Control Panel with Monitoring Dashboard (PC – UART) ..... | 4 |
| Block Diagram .....                                                      | 4 |
| Budget Estimation and Components List .....                              | 5 |

## Project Description

Greenhouse farming is a type of Controlled Environment Agriculture (CEA). It uses special buildings to control temperature, humidity, and sunlight. This helps crops grow as well as possible. **However**, most Greenhouses use very expensive equipment. Small-scale farmers cannot afford these high costs. Additionally, many industrial systems do not allow remote monitoring. This means farmers can only check on their crops while standing inside the greenhouse.

As a solution for above matters, specifically

- Hard affordability to small scale crop farms
- Inability to remotely monitor & control Greenhouse

We have come up with this project, **Affordable Remote Monitoring Capable Greenhouse Automation System**. Below are expected features and functionalities.

## Expected Functionalities

### 1. Soil Moisture Monitoring and Control

- \* System can analyze soil moisture in different sectors in greenhouse.
- \* System will analyze the soil moisture, get collective averages and check for the water levels of that soil, based on that data it can automate and monitor,
  - Soil moisture levels of the soil
  - If water levels are lower than in should be system can turn on the water supply
  - Whether the soil is good or bad for farming (Soil Health)

### 2. Air Temperature / Humidity Control

- \* System can measure air temperatures and humidity inside the greenhouse
- \* Based on the readings the system will determine when to trigger an exhaust fan to control temperature.

### 3. Greenhouse Efficiency (Sunlight) Monitoring

- \* System can detect low-high sunlight levels, and it can monitor photosynthesis levels to a dashboard to show when the greenhouse is most efficient.

#### 4. Water Tank / Reservoir Level Monitoring

- \* The system will monitor current water levels and notify on the dashboard when the tank/reservoirs on low water
- \* Automatic refill on low water

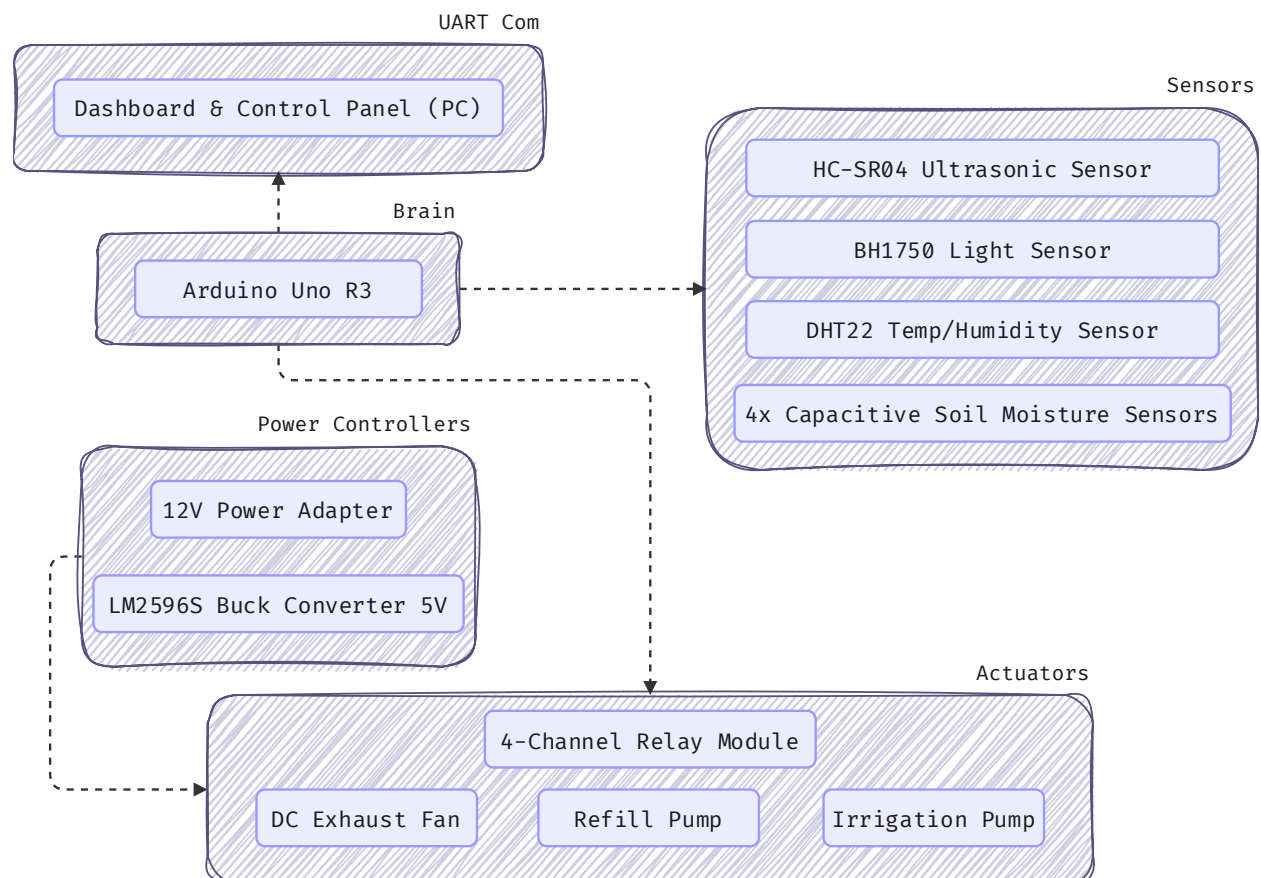
#### 5. Crop Watering based on Weather

- \* System uses a weather API + defined location (via Dashboard), to get geological weather information. If it is about to rain, the system will delay the watering systems to save water.

#### 6. Interactive Control Panel with Monitoring Dashboard (PC – UART)

- \* Ability to monitor Greenhouse in real time.
- \* Manually control individual components when necessary.
- \* Configure Greenhouse Thresholds (Moisture, Humidity) and Location (Weather API)

### Block Diagram



## Budget Estimation and Components List

| Component / Module                   | Qty | Source                                                                           | Price       |
|--------------------------------------|-----|----------------------------------------------------------------------------------|-------------|
| Arduino Uno R3                       | 1   | NIBM                                                                             | -           |
| 5V Mini Submersible Water Pump       | 2   | <a href="http://Duino.lk">Duino.lk</a>                                           | Rs. 430.00  |
| 1N4007 Rectifier Diode               | 2   | <a href="http://Duino.lk">Duino.lk</a>                                           | Rs. 30.00   |
| 5V Small DC Cooling Fan              | 1   | <a href="http://Tronic.lk">Tronic.lk</a>                                         | Rs. 190.00  |
| 5V 4-Channel Relay Module            | 1   | <a href="http://Duino.lk">Duino.lk</a>                                           | Rs. 560.00  |
| Capacitive Soil Moisture Sensor v1.2 | 4   | <a href="http://Duino.lk">Duino.lk</a>                                           | Rs. 880.00  |
| DHT22 / AM2302 Module                | 1   | <a href="http://Duino.lk">Duino.lk</a>                                           | Rs. 380.00  |
| BH1750 Light Sensor Module           | 1   | <a href="http://Tronic.lk">Tronic.lk</a>                                         | Rs. 450.00  |
| HC-SR04 Ultrasonic Sensor            | 1   | <a href="http://Tronic.lk">Tronic.lk</a>                                         | Rs. 260.00  |
| LM2596S Adjustable Buck Module       | 1   | <a href="http://Tronic.lk">Tronic.lk</a>                                         | Rs. 620.00  |
| 12V 2A Power Adapter & DC Pigtail    | 1   | <a href="http://Duino.lk">Duino.lk</a>                                           | Rs. 60.00   |
| Male & Female Header Pins (Strip)    | 2   | <a href="http://Duino.lk">Duino.lk</a><br><a href="http://Duino.lk">Duino.lk</a> | Rs. 230.00  |
| IDC Flat Wire Ribbon (1M)            | 4   | <a href="http://Tronic.lk">Tronic.lk</a>                                         | Rs. 400.00  |
| 2 Core Twin 24AWG Wire (1M)          | 4   | <a href="http://Duino.lk">Duino.lk</a>                                           | Rs. 200.00  |
| Total (Approx)                       |     |                                                                                  | Rs. 4690.00 |

\* Budget is Approximate + ([Heat Shrinking Tubes](#), [Ultrasonic Holder](#), [Wire Stripper](#))